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**3rd Sem / Auto, Mech, Prod, T&D, GE, CNC, CAD/  
CAM, Found. & Forg, Metallurgy, Print Making Tech.,  
Mech ( Ad. Manu. Tech.), Mech Engg ( Fabrication  
Tech), Mech. Engg. (Prod)**

**Subject:- Workshop Technology - I**

Time : 3Hrs.

M.M. : 100

### **SECTION-A**

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The distance from joint root to the toe of the weld is called
- a) Leg
  - b) Face
  - c) Effective throat
  - d) Actual throat
- Q.2 The electrodes used in Spot welding are NOT madeup of which material
- a) Only copper
  - b) Copper and tungstun
  - c) Copper and chromium
  - d) Copper and aluminum
- Q.3 Which of the following chemical compound, is the most common flux added to the charge in a cupola furnace?
- a) Coke
  - b) Methane
  - c) Carbon dioxide
  - d) Limestone

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## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Show the symbol of T-weld.
- Q.12 Name any two pattern allowances.
- Q.13 The skeleton pattern is filled with sand.(True/False)
- Q.14 \_\_\_\_\_ extrusion process is used for producing tooth pastes(impact/Tube)
- Q.15 In the foundry, the electric furnaces cannot operate for high temperatures (True/False)
- Q.16 What is the purpose of the oxygen stream in gas cutting?
- Q.17 Positive allowance means \_\_\_\_\_
- Q.18 Cause of “hot tears” in casting is due to rapid \_\_\_\_\_ (heating/cooling)
- Q.19 In pressure die casting, production rate is low. (True/False)
- Q.20 In a gas cutting torch, the oxygen and fuel gas are mixed inside \_\_\_\_\_ (flame/torch)

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the process of thread rolling.
- Q.22 Explain the BIS specification of arc welding electrodes.
- Q.23 Differentiate between Direct and Indirect extrusion.
- Q.24 Describe the working of cupola furnace.
- Q.25 Write short note on Injection moulding process.

- Q.26 Explain the working principle of centrifugal casting and its applications.
- Q.27 What are the various core materials?
- Q.28 Write short note on directional solidification and its importance.
- Q.29 Give the names of methods of testing of casting defects. Explain any one of them with diagram.
- Q.30 Write short note on principle of seam welding. Give its application.
- Q.31 Explain the Thermit welding process. Give its application.
- Q.32 What are various safety precautions in foundry shop.
- Q.33 Explain any 5 properties of moulding sand.
- Q.34 Differentiate between pit moulding and floor moulding.
- Q.35 Explain the difference in function of gate and riser with diagrams.

### **SECTION-D**

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the principle of operation, advantage disadvantage and application of TIG welding.
- Q.37 Classify various steps of patterns. Explain any 5 with diagrams.
- Q.38 Explain with diagrams any 5 rolling defects. Give their remedies.